

ARTICLE — ENGINEERING & DESIGN

Ingenia Magazine — Homes for the New Millennium (2001 + 2009 Retrospective)

Source: <https://www.ingenia.org.uk/articles/homes-for-the-new-millennium-new-technology-and-innovation-at-greenwich-millennium-village/>

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TITLE: Homes for the New Millennium: New Technology and Innovation at Greenwich Millennium Village

PUBLICATION: Ingenia Magazine, Issue 8, 2001

AUTHORS: Robert Hawley (Chairman, Taylor Woodrow PLC) and Richard Hodgkinson (Innovation Consultant)

SUMMARY:

GMV is an innovative mixed-use residential development on the eastern side of Greenwich Peninsula adjacent

to the River Thames. 1,377 homes, 20% affordable/social housing integrated with private housing. Su

pplemented by commercial and retail space. All homes linked through an IT network with internal data

cabling.

Developer: Taylor Woodrow Capital Developments and Countryside Properties joint venture (Greenwich Millennium Village Ltd).

Project value: approximately £250 million to be built over 6 years.

Construction commenced: late 1999. First houses occupied: December 2000.

TECHNICAL INNOVATION TARGETS:

- Construction waste: reduce by 50%
- Primary energy: reduce by 80% (includes CHP, improved insulation, better heating controls – achieves 65% reduction in practice)
- Water consumption: targeted reduction of up to 30%
- Construction costs and project duration: targeted reductions of up to 30%

ENERGY:

- GMV was the first major UK private housing development to incorporate Combined Heat and Power (CHP)
- CHP facilities generate electricity locally; associated heat serves hot water and heating demand
- Village-wide community heating system – metered at each dwelling
- CO2 reduction of ~67% versus all-electric benchmark
- Phase 2 energy centre sized to meet total energy demand for 550 dwellings in that phase

STRUCTURE:

- Concrete framed construction for high-rise apartments
- Partial use of cement replacement concrete; wide-scale use for high-rise frames proved unviable due to early age strength gain constraints
- Off-site construction attempted but challenging to achieve cost-effectively at targeted standards
- International sourcing of products; emphasis on standardisation across whole project

ENVIRONMENTAL DESIGN:

- Passive-solar orientation
- Large south-facing glazing
- External shading via balconies/canopies
- High-grade insulation
- Efficient appliances
- Car parking restricted and sited away from properties
- Secure bicycle storage
- Ecology park and landscaped public spaces

RETROSPECTIVE (Ingenia Issue 41, c.2009 – "Millennium Homes Revisited"):

- EcoHomes 'Excellent' rating achieved
- Lessons learned from running CHP system for 8+ years
- Two different CHP solutions incorporated for phases 1 and 2
- CO2 reduction confirmed at 67% vs all-electric benchmark
- Note: important lessons learned over the 8 years the system had been running

NOTE ON STRUCTURE DETAIL (from Ingenia revisit):

"To create the high residential density for Greenwich Millennium Village, concrete framed construction was used for the high rise apartments."

Retrospective article URL:

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